Extending n-sequences

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Classical Multiplier Sequences

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Let $\{\gamma_k\}_{k=0}^{\infty}$ be a sequence of real numbers. We call $\{\gamma_k\}$ a *classical multiplier sequence* if $\sum_{k=0}^{n} \gamma_k a_k x^k$ is hyperbolic whenever $\sum_{k=0}^{n} a_k x^k$ is hyperbolic.
We use *CMS* to denote the set of all classical multiplier sequences.

Proposition

A polynomial of the form $f(x) = \sum_{k=0}^{m} a_k x^k + \sum_{k=m+3}^{n} a_k x^k$ *is not hyperbolic.*

Algebraic Characterization

Theorem (Pólya-Schur)

Let $\{\gamma_k\}_{k=0}^{n} \in \mathbb{R}^{n+1}$. *Then the following are equivalent:*

- $\sum_{k=0}^{m} a_k \gamma_k x^k$ *is hyperbolic whenever* $\sum_{k=0}^{m} a_k x^k$ *is hyperbolic.*
- $\sum_{k=0}^{n} \binom{n}{k} \gamma_k x^k$ *has only real zeroes of the same sign or zero.*

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Definition

If the conditions of the above theorem are satisfied, then we say that $\{\gamma_k\}_{k=0}^n$ an *n-sequence* and that $\sum_{k=0}^n \binom{n}{k} \gamma_k x^k$ is the *Jensen polynomial* associated with $\{\gamma_k\}_{k=0}^n$.

We let CMS_n denote the set of all n-sequences and $\mathcal{H}_n$ denote the set of all polynomials over $\mathbb{R}$ of degree at most n which have only real zeroes of the same sign or zero.

Topological Equivalence

Proposition

Let

$$\phi\left(\sum_{k=0}^{n}\binom{n}{k}\gamma_k x^k\right) = \{\gamma_k\}_{k=0}^{n}$$

Then $\phi : \mathcal{H}_n \to CMS_n$ *is a homeomorphism for each* $n \in \mathbb{N}$.

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Remark

For each $n \in \mathbb{N}$, *we interpret* $\mathcal{H}_n$ *as a topological subspace of* $\mathbb{R}_n[x]$ *with the compact convergence topology. Similarly, we interpret* CMS_n *as a topological subspace of* $\mathbb{R}^{n+1}$ *with the Euclidean metric.*

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For $n \geq m$, $\{\gamma_k\}_{k=0}^{m} \in CMS_m$, and $\{\delta_k\}_{k=0}^{n} \in CMS_n$, we say that $\{\delta_k\}_{k=0}^{n}$ is an *extension* of $\{\gamma_k\}_{k=0}^{m}$ if $\gamma_k = \delta_k$ for $0 \leq k \leq m$.

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Note that, for any m-sequence $\{\gamma_k\}_{k=0}^m$, and any $n < m$, $\{\gamma_k\}_{k=0}^n$ is an n-sequence.

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Question

For any $m > n$, can every n-sequence be extended to an m-sequence? If not, can we characterize which ones can be extended, and how?

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- *If $\{\gamma_k\}_{k=0}^{n}$ is a sequence of real numbers with $\gamma_0 = 1$ and $\gamma_{k-1}^2 \geq 4(1-1/k)\gamma_k\gamma_{k-2}$ for $k \in \{2, 3, \ldots, n\}$, then $\{\gamma_k\}_{k=0}^{n}$ is an n-sequence which is extendable to CMS.*

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Example

Let $f(x) = 2x^5 + 3x^4 + 9x^2 - 3x + 1$. Then
$f^*(x) = x^5 - 3x^4 + 9x^3 + 3x + 2$.

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Proposition

Reverse preserves hyperbolicity of polynomials and the sign and multiplicity of their nonzero zeros.

Integral Representation

Proposition

If g_n and g_{n+1} are the Jensen polynomials associated with $\{\gamma_k\}_{k=0}^{n}$ and $\{\gamma_k\}_{k=0}^{n+1}$, respectively, then

$$g_{n+1}^{*'}(x) = (n+1)g_n^*(x)$$

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Corollary

If g_n is a Jensen polynomial, then the associated n-sequence $\{\gamma_k\}_{k=0}^{n}$ is extendable to CMS_{n+1} if and only if there exists an $a \in \mathbb{R}$ such that

$$f(x) = (n+1)\int_a^x g_n^*(t)dt \in \mathcal{H}_{n+1}$$

In this case, $\{\gamma_k\}_{k=0}^{n} \cup \{a\}$ is an $(n+1)$-sequence, with the associated Jensen polynomial given by $g_{n+1}(x) = f^*(x)$.

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The following hold:

- *For any $2 \leq m < n$ and $a \neq 0$, the n-sequence associate with $g_n(x) = (x+a)^m$ cannot be extended to an $(n+1)$-sequence.*
- *For each integer $n \geq 3$, there exists a n-sequence of nonzero terms which cannot be extended to an $(n+1)$-sequence.*

Conjecture

Consider $q(x) = (x+a)^m p(x) \in \mathcal{H}_n$, where $p \in \mathbb{R}_n[x]$, $\deg q < n$, $a \neq 0$, and $m \geq 2$. Then the n-sequence associated with q cannot be extended to an $(n+1)$-sequence.

An Equivalent Question

Take $g_n(x) = (x+a)^2 q(x)$ for some $a \in \mathbb{R}_{\neq 0}, q(x) \in \mathcal{H}_{n-3}$, where the roots of $q(x)$ are nonzero and have the same sign as $-a$.

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Proposition

$g_{n+1}(x)$ has roots of the same sign (or zero) if and only if $g^*_{n+1}(x)$ has roots of the same sign (or zero).

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Proposition

$g_{n+1}(x)$ has roots of the same sign (or zero) if and only if $g^*_{n+1}(x)$ has roots of the same sign (or zero).

Our conjecture is then equivalent to showing that:

$$g^*_{n+1}(x) = (n+1)\int_0^x t^{n-m}(at+1)^2 q^*(x)dt$$

does not have real roots of the same sign.

A Conjecture Regarding Jensen Polynomials

Conjecture

Consider an n-sequence, $\{\gamma_k\}_{k=0}^{n}$ *whose last term is zero, and let* $g_n(x)$ *be its associated nth Jensen Polynomial. If* $g_n(a) = 0$ *for some nonzero* $a \in \mathbb{R}$*, then* $g_{n+1}(a) \neq 0$*.*

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By Taylor's theorem, we have

$$g_n(x) = 0 + g_n'(a)(x-a) + \frac{g''(a)(x-a)^2}{2!} + \ldots + \frac{g_n^{(m)}(a)(x-a)^m}{m!}$$

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Using the integral representation, we get the following:

$$g_{n+1}^{*}\left(\frac{1}{a}\right) = \sum_{k=0}^{m} \frac{(-1)^{n-k+1} g_n^{(k)}(a) k!}{(n+1)!}$$

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Then, for $C \neq 0$, we have

$$g^*_{n+1}\left(-\frac{1}{a}\right) = C + \frac{g^{*(3)}_{n+1}\left(-\frac{1}{a}\right)\left(x+\frac{1}{a}\right)^3}{3!} + \ldots + \frac{g^{*(m)}_{n+1}\left(-\frac{1}{a}\right)\left(x+\frac{1}{a}\right)^m}{m!}$$

Partial Results and Further Research

Proposition (H-P)

Let $\{a_k\}$ *denote the zeros of the polynomial* $p(x)$ *in the above conjecture. If, in addition to the above conditions, we assume that* $|a| \geq |a_k|$ *or* $|a| \leq |a_k|$ *for all* $1 \leq k \leq \deg p$*, then the result follows.*

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Under what conditions will an n-sequence of nonzero terms be non-extendable to an $(n+1)$-sequence? Is the condition due to Craven and Csordas necessary, or can it be strengthened?

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